OFFICIAL OPERATIONS MANUAL • V2.4 WIN-NATIVE

NETSPECTRE

Real-Time Windows Network Traffic Intelligence, Threat Surveillance & Multi-Dimensional Visualization Platform

CORE PLATFORM PURPOSE

NETSPECTRE monitors all network traffic entering and leaving your Windows 10/11 computer, correlating low-level socket activity with exact executable processes, remote server identities, cloud infrastructures, and transport encryption states in real time.

APPLICATION EDITION
NETSPECTRE v2.4 (Production Win64)

TELEMETRY ENGINE
C# Native P/Invoke (iphlpapi.dll)

USER INTERFACE
Electron 32 • Vite • Three.js WebGL

OPERATING SYSTEM
Windows 10 / Windows 11 (64-bit)

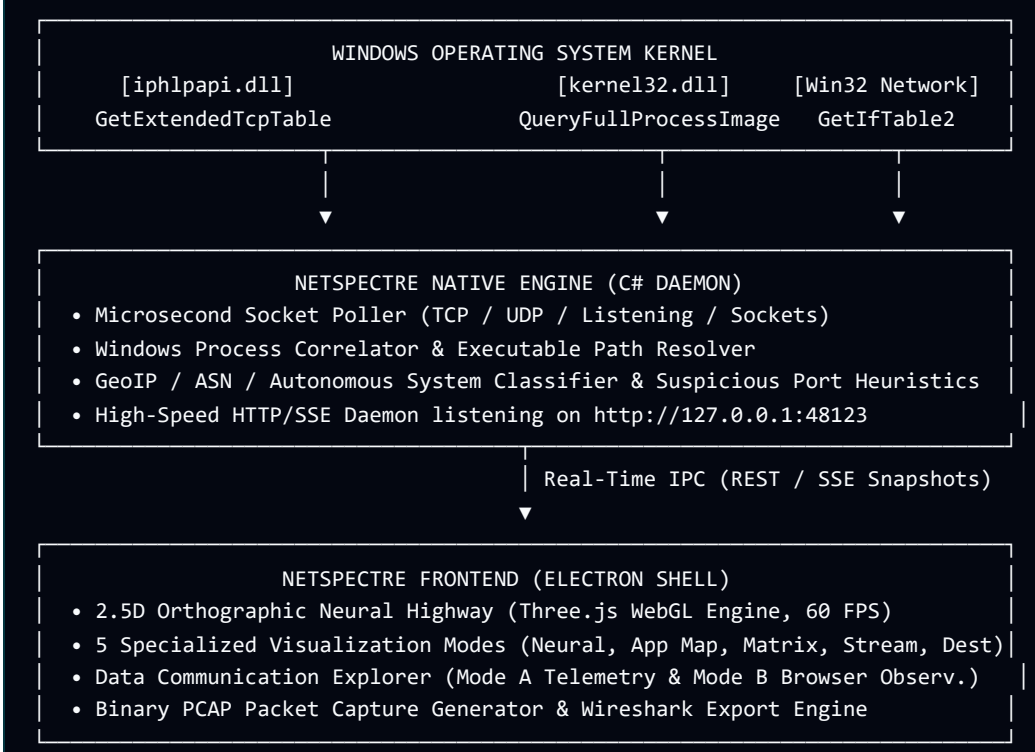
1. EXECUTIVE OVERVIEW & CAPABILITIES

NETSPECTRE bridges the gap between low-level packet sniffers (like Wireshark) and per-application firewalls (like GlassWire). Traditional tools present overwhelming raw hex dumps or oversimplified numbers. NETSPECTRE combines deep Windows kernel-level socket awareness with a futuristic 2.5D neural visualization command center.

Primary Business & Security Questions Answered

QUESTION	WHAT NETSPECTRE DISCLOSES
What data enters my PC?	Real-time download throughput (B/s, KB/s, MB/s), cumulative downloaded volume, and individual inbound frame packet sizes.
What data leaves my PC?	Real-time upload speed, cumulative sent bytes, and outbound request frame sizes with directional tracking.
Which application is talking?	Exact process name (`chrome.exe` , `Unity.exe` , `msedge.exe` , `svchost.exe`), Process ID (PID), and physical binary path on disk.
Where is it sending data?	Observed remote IPv4/IPv6 addresses, resolved domain names (FQDN), and remote destination ports (443, 53, 80, 8080).
Who owns the remote server?	Hosting organization attribution (Cloudflare, Microsoft Azure, Google Cloud, AWS), Autonomous System Number (ASN), and country geolocation flags.
Which protocol is used?	Transport protocol (TCP / UDP) and application classification (TLS 1.3 / HTTPS, QUIC / HTTP3, DNS 53, mDNS 5353).
Is the data encrypted?	Full cryptographic transparency: clearly labels encrypted payloads (TLS 1.3/QUIC) vs unencrypted traffic without deceptive claims.

2. SYSTEM ARCHITECTURE & DATA FLOW



> **100% REAL WINDOWS TELEMETRY GUARANTEE**
>
> NETSPECTRE never generates simulated or fictional traffic. Every node, curved link, packet gauge, and particle represents genuine active socket telemetry intercepted from your physical Windows network adapter.

3. QUICK START & LAUNCH INSTRUCTIONS

3.1 System Prerequisites

- **Operating System:** Windows 10 or Windows 11 (64-bit edition).
- **Runtime:** Microsoft .NET Framework 4.5 or higher (pre-installed on all modern Windows systems).
- **Node.js:** Node.js v18+ with npm (for desktop frontend rendering).

3.2 One-Click Desktop Launch

To start the application, navigate to the application root directory and double-click the launcher:

```
launch.bat
```

The launcher performs the following automated verification sequence:

1. Checks if `src-engine\NetSpectreEngine.exe` exists; if missing, automatically compiles it using the native 64-bit C# compiler.
2. Verifies that the telemetry service is responding on `http://127.0.0.1:48123/api/status`.

3. Spawns the Electron desktop command center with hardware acceleration enabled.

3.3 Command-Line Startup Options

COMMAND	PURPOSE
`launch.bat`	Primary one-click user launcher for the complete suite.
`npm.cmd run electron`	Launches the desktop Electron GUI in production mode.
`npm.cmd run build:ui`	Recompiles Vite, TypeScript, Tailwind CSS, and WebGL assets.
`.\src-engine\NetSpectreEngine.exe --port 48123`	Runs the native C# telemetry daemon as a standalone console service.

4. USER INTERFACE & WORKSPACE NAVIGATION

1. TOP TELEMETRY HUD BAR

- **Downlink & Uplink Speeds:** Instantaneous digital gauges (KB/s, MB/s).
- **Active Flows Badge:** Sockets count and active application tally.
- **Hardware Adapter Badge:** Displays your physical NIC (e.g. *Wi-Fi TP-Link*).
- **Global Search:** Filters nodes by app, domain, IP, or port.
- **Pause / Play Button:** Freezes visualization to inspect a burst.
- **PCAP Export:** One-click download of binary packet captures.

2. SURVEILLANCE MODULES (LEFT)

- **Neural Universe:** The main 2.5D animated command highway.
- **Data Explorer:** Microsecond frame inspector & payload disclosure.
- **Applications:** Per-process socket and volume catalog.
- **Connections:** Comprehensive tabular TCP/UDP socket matrix.
- **Packet Explorer:** Real-time frame waterfall and packet analysis.
- **Destinations:** Global geographic & ASN intelligence.
- **Time Travel:** Historical playback scrubber and replay controls.
- **Security & Threats:** Automated alert engine for suspicious sockets.

3. BOTTOM EVENT STREAM & RIGHT CONTEXT DRAWER

Real-Time Event Stream (Bottom): Continuously logs socket lifecycle events (`CONNECT` , `DISCONNECT`) with exact timestamps, process attribution, and remote endpoints. Click "*Click to collapse*" to expand or minimize.

Context Inspector Drawer (Right): Clicking on any node or socket opens a deep forensic drawer showing remote ASN, byte counters, packet tallies, duration, copyable IPs, and suspicious heuristics.

5. THE 2.5D NEURAL UNIVERSE & 5-COLUMN HIGHWAY

The **Neural Universe** is the centerpiece of NETSPECTRE. It eliminates confusing 3D wireframe tangles in favor of an organized, 2.5D orthographic pipeline flowing cleanly from left to right.

The 5 Symmetrical Highway Columns

COLUMN	COORDINATES & VISUALS	CONTENTS & NETWORK TELEMETRY
1. Local System	X = -380 Cyan Core (#00f3ff)	Represents your Windows PC. Displays computer name, local LAN IP address (e.g. `192.168.1.112`), active hardware adapter, and total up/down speeds.
2. Applications	X = -190 Process Color Badges	Running Windows processes sorted by active bandwidth. Shows process names (`chrome`, `Unity`, `msedge`, `svchost`), PID, and active sockets.
3. Protocol Hubs	X = 0 Sky Blue / Indigo	Consolidated transport protocol hubs: `TCP 443 (TLS)`, `UDP 443 (QUIC)`, `UDP 53 (DNS)`, `TCP 80 (HTTP)`, and `UDP 5353 (mDNS)`.
4. Destinations	X = +190 GeoIP Badges & Flags	External servers contacted by your computer. Displays country flags, remote IP addresses, domain names, and destination port numbers.
5. Service Groups	X = +380 Purple Cluster Nodes	Infrastructure routing attribution: aggregates endpoints into known Autonomous Systems such as *Cloudflare*, *Microsoft Azure*, *Google Cloud*, *Amazon AWS*, and *RFC1918 Private*.

Particle Flow Dynamics & Color Coding

AMBER PARTICLES — OUTGOING DATA (UPLINK)

Particles moving from left to right represent upload traffic originating from your computer towards the remote server. Speed and density scale directly with measured outbound throughput.

CYAN PARTICLES — INCOMING DATA (DOWNLINK)

Particles moving from right to left represent download traffic returning from external servers into your local applications. High particle velocities indicate active streaming or file downloads.

Canvas Interactive Controls

- **Pan View:** Left-click and drag anywhere on the canvas to pan across the highway.
- **Zoom In / Out:** Scroll the mouse wheel, or use the `[+]` and `[-]` buttons in the bottom-right corner.
- **Reset View:** Click the `RESET VIEW` button to restore the camera to the default centered perspective.
- **Node Hover Tooltip:** Hover over any node to reveal an instant HUD card detailing real-time speeds, transferred bytes, and PID.
- **Path Highlighting:** Click any node to highlight all incoming and outgoing Bezier paths connected to it in bright white.

6. THE 5 SPECIALIZED VISUALIZATION MODES

Switch between the 5 visualization modes using the floating pill bar at the top of the CommandCenter:

Mode 1: Neural Flow (2.5D Data Highway)

The primary real-time WebGL data-flow universe described in Section 5. Best for continuous situational awareness and observing directional flow dynamics across all 5 architectural layers.

Mode 2: Application Map (Bipartite Process-to-Endpoint Matrix)

Presents a clean, two-column bipartite map linking running applications on the left directly to contacted endpoints on the right.

- **Process Isolation:** Clicking any application (e.g. `chrome.exe`) instantly filters the right-hand column to display *only* the remote servers contacted by that specific process.
- **Quick Search:** Includes a dedicated application search bar to locate specific processes instantly.

Mode 3: Live Data Stream (Traffic Burst Waterfall)

Visualizes raw packet bursts in real time as a continuous microsecond waterfall feed:

- **Directional Badges:** Green `RECV` vs Amber `SEND` indicators with packet lengths in bytes.
- **Burst Gauges:** Dynamic horizontal glowing bars representing relative frame volume.
- **Direction Filters:** Toggle between `ALL`, `INCOMING`, and `OUTGOING` traffic.
- **Pause Stream:** Freeze the live stream to inspect a high-frequency burst.

Mode 4: Destination Radar (Geographic & ASN Intelligence)

Groups your network activity by physical geography and Autonomous System routing:

- **Observed Regions:** Shows all active countries sorted by socket volume with national flags and byte tallies.
- **Region Isolation:** Click any country badge (e.g. us United States) to view all active sockets connected to that specific territory.

Mode 5: Protocol Matrix (Application × Protocol Heatmap)

A high-density tabular heatmap cross-referencing running processes against network protocols:

- Columns cover `TCP 443 (TLS)`, `UDP 443 (QUIC)`, `UDP 53 (DNS)`, `TCP 80 (HTTP)`, `UDP 5353 (mDNS)`, and `Other Sockets`.
- Heat cells highlight volume in cyan, with socket counts and byte counters. Click any cell to inspect the underlying socket.

> **TOP CONTROL PILL FILTERS**
>
> The top bar also provides real-time protocol filters (`ALL`, `TCP`, `UDP`), traffic filters (`ALL`, `ACTIVE`, `PUBLIC`, `PRIVATE`, `SUSPICIOUS`), and an `Active Only` toggle to hide idle sockets.

7. DATA COMMUNICATION EXPLORER & PAYLOAD DISCLOSURE

The **Data Communication Explorer** (accessible via the left sidebar) provides forensic timeline inspection of individual socket flows.

7.1 Chronological Frame Sequence (Mode A)

Select any active socket flow from the left-hand column to view its chronological request and response frames:

```
[18:54:22.000] OUTGOING REQUEST  (1500 Bytes)  192.168.1.112:49676 → 4.213.25.241:443  [ACK]
[18:54:22.000] INCOMING RESPONSE ( 128 Bytes)  4.213.25.241:443 → 192.168.1.112:49676  [ACK, PSH]
[18:54:22.792] OUTGOING REQUEST  (1500 Bytes)  192.168.1.112:49676 → 4.213.25.241:443  [ACK]
[18:54:22.792] INCOMING RESPONSE ( 151 Bytes)  4.213.25.241:443 → 192.168.1.112:49676  [ACK, PSH]
```

7.2 Cryptographic Transparency Notice

> **TRANSPORT ENCRYPTION (TLS 1.3 / QUIC) REALITY**
>
> Modern HTTPS and QUIC traffic is encrypted end-to-end between the client and remote endpoint using Ephemeral Diffie-Hellman keys (ECDHE). NETSPECTRE adheres strictly to genuine technical accuracy: it truthfully captures packet sizes, arrival intervals, and endpoint addresses without deceptively claiming to read encrypted passwords or private tokens on the wire.

7.3 Mode B: Optional Browser Request Observability

For browser sessions (such as ChatGPT, Google, or YouTube), switch the top toggle to **Mode B: Browser Observability**.

- Click **ENABLE USER CONSENT** to authorize request-level observability.
- Displays higher-level HTTP request URLs (e.g. `POST https://chatgpt.com/backend-api/conversation`), HTTP status codes (`200 OK`), content types (`application/json`), and transfer sizes.

8. WIRESHARK INTEGRATION & PCAP EXPORT

NETSPECTRE includes an integrated packet capture generator for deep forensic analysis:

1. Click the **PCAP** button in the top navigation bar.
2. NETSPECTRE exports a standard binary PCAP file (magic number `0xa1b2c3d4`) containing captured Ethernet and IP frames.
3. Double-click the exported `.pcap` file to open it directly in **Wireshark**.

> **DEVELOPER TIP: TLS PAYLOAD DECRYPTION IN WIRESHARK**
>
> If you wish to view decrypted HTTP payloads in Wireshark for your own browser sessions: set the Windows environment variable `SSLKEYLOGFILE=C:\sslkeys.log`, restart your browser, and point Wireshark's TLS preferences (*Edit → Preferences → Protocols → TLS → (Pre)-Master-Secret log filename*) to that file.

9. COMPLETE MODULE REFERENCE GUIDE

MODULE	SIDEBAR ROUTE	PRIMARY FUNCTIONALITY & USE CASE
Neural Universe	`Neural Universe`	Interactive 2.5D visual command center with 5 columns and 5 visualization modes.
Data Explorer	`Data Explorer`	Microsecond frame sequence timeline, TLS transparency disclosures, and Mode B browser request telemetry.
Applications	`Applications`	Comprehensive per-application catalog showing total upload/download volume, active socket count, and contacted domains.
Connections	`Connections`	Detailed tabular matrix of every active socket: protocol, local address, local port, remote address, remote port, and TCP state.
Packet Explorer	`Packet Explorer`	Real-time packet stream breakdown with direction indicators, flags, and size distribution.
Destinations	`Destinations`	Global endpoint inventory with GeoIP mapping, ASN attribution, and country grouping.
Time Travel	`Time Travel`	Historical playback module. Scrubber timeline allows replaying past network bursts.
Analytics	`Analytics`	Time-series bandwidth graphs, protocol distribution donut charts, and transfer totals.
Security & Threats	`Security & Threats`	Automated threat detection engine flagging abnormal ports, unauthorized background listeners, and suspicious telemetry.
Settings	`Settings`	Configure engine IPC port (default 48123), visual rendering quality, and particle density.

10. TROUBLESHOOTING & FREQUENTLY ASKED QUESTIONS

Q1: Why does PowerShell show a SecurityError regarding npm.ps1?

Windows PowerShell disables `.ps1` script execution by default. Always launch the application using `launch.bat` or call `npm.cmd` directly instead of `npm`.

Q2: Can NETSPECTRE decrypt HTTPS passwords or banking details?

No. In accordance with zero-trust security and technical honesty, NETSPECTRE never compromises cryptographic security. It monitors network dynamics, socket metadata, and packet timings without intercepting private credentials.

Q3: How do I change the engine port if 48123 is occupied?

Launch the engine with the custom port flag: `NetSpectreEngine.exe --port 50123`, and update the port in NETSPECTRE Settings or store configuration.